

Epidemiology of Diseases of Public Health Significance

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Disease of public health significance

- Reportable Diseases are now referred to as "Diseases of Public Health Significance".
- Timely reporting of communicable diseases is essential for their control
- In Ethiopia, there are 22 reportable priority diseases or conditions, which are included in the IDSR system

List of priority disease under surveillance reporting in Ethiopia

Epidemic-prone diseases

1. Cholera
2. Diarrhea with blood (Shigellosis)
3. Measles
4. Meningitis
5. Plague
6. Viral hemorrhagic fevers
7. Yellow fever
8. Typhoid fever
9. Relapsing fever
10. Epidemic typhus
11. Malaria

Diseases targeted for eradication

12. Acute flaccid paralysis (Polio)
13. Dracunculiasis (Guinea worm)
14. Leprosy
15. Neonatal Tetanus

Other diseases of public health importance

16. Pneumonia in children
17. Diarrhea in children
18. New AIDS cases
19. Onchocerciasis
20. Sexually transmitted diseases
21. Tuberculosis

Why prioritize?

- Priority setting is necessary to **ensure that both planning and resource allocation** are rational, explicit and transparent
- **Surveillance systems** are usually developed over time, with new diseases being added and few removed.
- This often results in a long list of diseases for surveillance, impairing the ability of the system to perform efficiently.
- In many surveillance systems data are collected which never result in public health action,
 - ✓ whereas new threats are considered insufficiently or not at all.

Priority cont...

Public health risks change over time therefore priorities should be reviewed periodically. Because;

- Some diseases are **no longer a national public health threat** and yet they remain on the national list of mandatory reportable diseases
- **Diseases evolve** as antigenic variants expand their host or geographical range or as antibiotic resistance develops.
- **Advanced technology** and improved knowledge impact on surveillance strategies and methods.
- The introduction of **new vaccines and control programmes** alters the surveillance objective and leads to different surveillance methods.
- As demographic characteristics of **populations change** due to lower birth rates and longer life expectancy, new risks may be expressed.

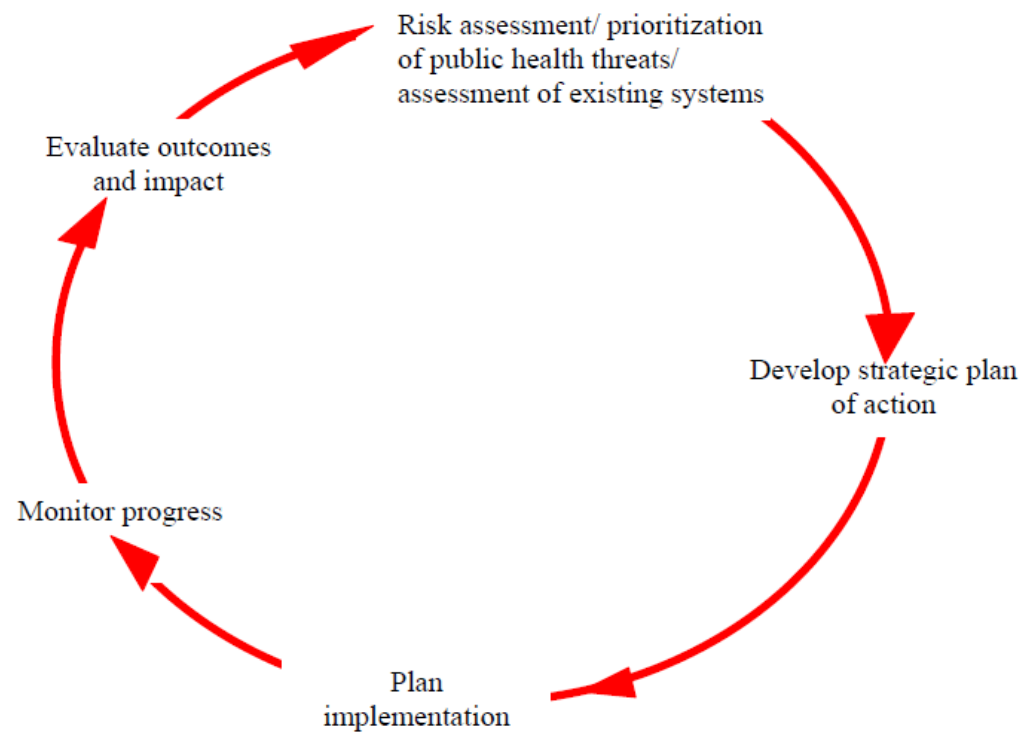
In order to capture data on the most important diseases and conditions and not to overstress the surveillance system, priorities have to be set.

When to prioritize?

- All countries should review their list of priority diseases for surveillance periodically to see if it reflects current needs.
- Prioritization only makes sense if it happens within the right context;
 - Political endorsement of the process and willingness to accept the results of the exercise are prerequisites.

Good opportunities to revise priority diseases are:

- As an initial step to strengthen national surveillance, either before or immediately after the assessment of the existing system
- As part of the process of evaluation of surveillance functions;
- When there is a change in the national health policy or in the International Health Regulations;
- Following an outbreak, to take advantage of lessons learned and public pressure and political will to change;
- Whenever there is a new threat, or a new disease. Case of COVID19.



Cycle illustrating surveillance systems strengthening activities

How to prioritize?

- Prioritizing diseases for surveillance is not an easy process.
- It involves complex value judgments,
 - such as the relative importance of early detection of a highly infectious disease compared with monitoring endemic, common, but less severe diseases.
- Ideally, prioritization should be based on scientific evidence
- In situations where there is insufficient, inadequate, or contradictory scientific information or it is non-existent, consensus methods, such as the **Delphi method**, are valid approaches.
- The **Delphi method provides** a structure and process to harness the insight of appropriate experts to enable decisions to be made.
 - It avoids personal and political influence,
 - allows individuals to change their opinion in the light of the group's response.
 - Measuring the degree of agreement helps to move to a consensus.

Prioritization process using the Delphi method consists of:

- Formulation of a list of diseases, and criteria to include/exclude diseases for surveillance, by a steering committee;
- Discussion of the proposed criteria and disease list by participants in the prioritization exercise;
- Formulation of a questionnaire in which participants are asked to score the diseases against the criteria;
- Expression of individual opinion of participants through scoring the diseases against the criteria;
- Collation and summary (using statistical parameters) of the scoring, and assessment of agreement;
- Feedback of the individual and group ranking to the participants;
- Discussion of the results

Some Disease of Public Health Importance

Acute respiratory illness (ARI)

- About 90% of all ARI among under-five children deaths were reported in the Sub-Saharan Africa and Southeast Asia.
- Globally, more than 12 million children with severe ARI were admitted to hospitals every year.
- In Ethiopia, about 3.4 million children suffer from ARI annually and Ethiopia is among the top 15 countries with highest burdens of ARI .
- It accounts for 18% of all deaths, and **kills over 40,000 under-five children every year.**

ARI Cont...

- Acute respiratory infection (ARI), and particularly pneumonia, is one of leading causes of morbidity and mortality
 - ✓ That accounts for 18% of deaths (WHO and UNICEF 2013).
- Ethiopian Demographic and Health Survey (EDHS) data 2016.
 - 7% of children under age 5 had symptoms of ARI in the 2 weeks before the survey.
 - Treatment was sought for three out of 10 children and only 3% of these children received treatment on the same or next day
 - ARI among under-five children had spatial variations across the country (Moran's I: 0.34, $p < 0.001$).

Predictors of under-five acute respiratory infection.

- History of diarrhea, age, working mothers, secondary school education, and stunting.

Frequency Distributions of Childhood Morbidities across Regions in Ethiopia

Region	Morbidity in the Last Two Weeks				
	Cough N (%)	Fever N (%)	Shortness of Breath N (%)	Chest N (%)	ARI Combined N (%)
Tigray	302 (30.4)	234 (23.6)	161 (16.2)	161 (16.2)	167 (16.8)
Afar	163 (16.8)	152 (15.6)	72 (7.4)	71 (7.3)	95 (9.8)
Amhara	158 (17.0)	113 (12.2)	109 (11.7)	107 (11.5)	77 (8.3)
Oromia	347 (23.2)	221 (14.8)	224 (15.0)	224 (15.0)	157 (10.5)
Somali	125 (8.9)	114 (8.1)	66 (4.7)	66 (4.7)	81 (5.8)
Benishangul	41 (5.0)	59 (7.2)	21 (2.6)	20 (2.5)	26 (3.2)
SNNPR	211 (17.5)	188 (15.6)	121 (10.0)	120 (10.0)	112 (9.3)
Gambela	55 (8.2)	95 (14.2)	40 (6.0)	40 (6.0)	41 (6.1)
Harari	41 (7.3)	53 (9.4)	17 (3.0)	17 (3.0)	30 (5.3)
Addis Ababa	68 (15.2)	66 (14.8)	29 (6.5)	27 (6.0)	51 (11.4)
Dire Dawa	75 (14.5)	59 (11.4)	31 (6.0)	31 (6.0)	48 (9.3)
Total	1586 (15.9)	1354 (13.5)	891 (8.9)	884 (8.8)	885 (8.8)

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Disparity of ARI

- The incidence of ARI among under-five children varies among countries at large and regions in particular.
- These differences can be attributed to child, mother, environment related factors, and co-morbid diseases like: measles, diarrhoea, and malaria

Human immune deficiency Virus (HIV)

- HIV continues to be a major global public health issue, having claimed 36.3 million [27.2–47.8million] lives so far.
- There were an estimated 37.7 million [30.2–45.1 million] people living with HIV at the end of 2020
 - ✓ **Over two thirds of whom** (25.4 million) are in the WHO African Region.
- In 2020, 680 000 [480 000–1.0 million] people died from HIV-related causes and 1.5 million [1.0–2.0million] people acquired HIV
- Globally, 27.5 million [26.5–27.7 million] people living with HIV were receiving ART in 2020.
 - ✓ This equates to a global ART coverage rate of **73% [56–88%]**.

HIV/ADIS cont...

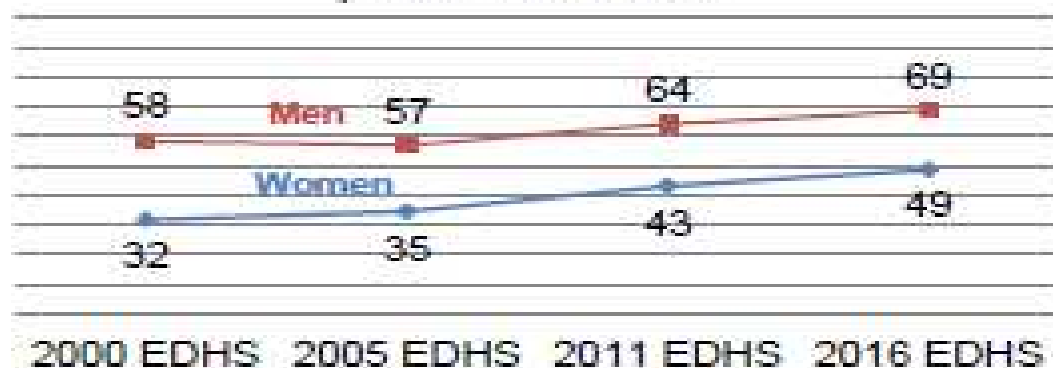
- In Ethiopia, Progress towards the 2020 90-90-90 targets for HIV was suboptimal for the first 90
 - ✓ with only 78.5% of estimated people living with HIV (PLHIV) knowing their HIV status.
- Achievements were better for the second and third 90's:
 - ✓ 90% of PLHIVs who know their status were on ART and
 - ✓ 91% of those on ART had viral suppression (MOH, 2019/20).
- About 90% of the annual incident HIV infections in children under 15 are still due to mother-to-child transmission (13.4%, according to UNAIDS 2019).

Table 1. Summary of performance for HSTP-I key performance indicators

Domain	Indicator	2015 baseline	HSTP-I Target	Achievement
Health status	Life expectancy at birth (in years)	64	69	65.5
	Maternal morality ratio	420	199	401
	Under five mortality (per 1,000 live births)	64	30	59
	Infant mortality (per 1,000 live births)	44	20	47
	Neonatal morality (per 1,000 live births)	28	10	33
	Stunting (under five years old)	40	26	37
	Wasting (under five years old)	9	4.9	7
	HIV incidence rate (per 1,000)	0.03	0.25	0.16
	TB incidence rate (per 100,000)	224	156	151
	TB mortality rate (per 100,000)	32	17	22
	Malaria death rate (per 100,000)		0.6	0.32

Figure 12.1 Knowledge of HIV prevention methods

Percentage of women and men age 15-49 who have knowledge of HIV prevention methods*

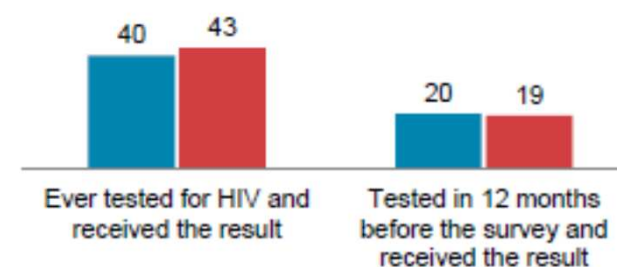


* Percentage who, in response to prompted question, say that people can reduce the risk of getting HIV by using condoms every time they have sexual intercourse, and by having one sex partner who is not infected and has no other partners.

Figure 12.6 HIV testing

Percentage of women and men age 15-49

■ Women ■ Men

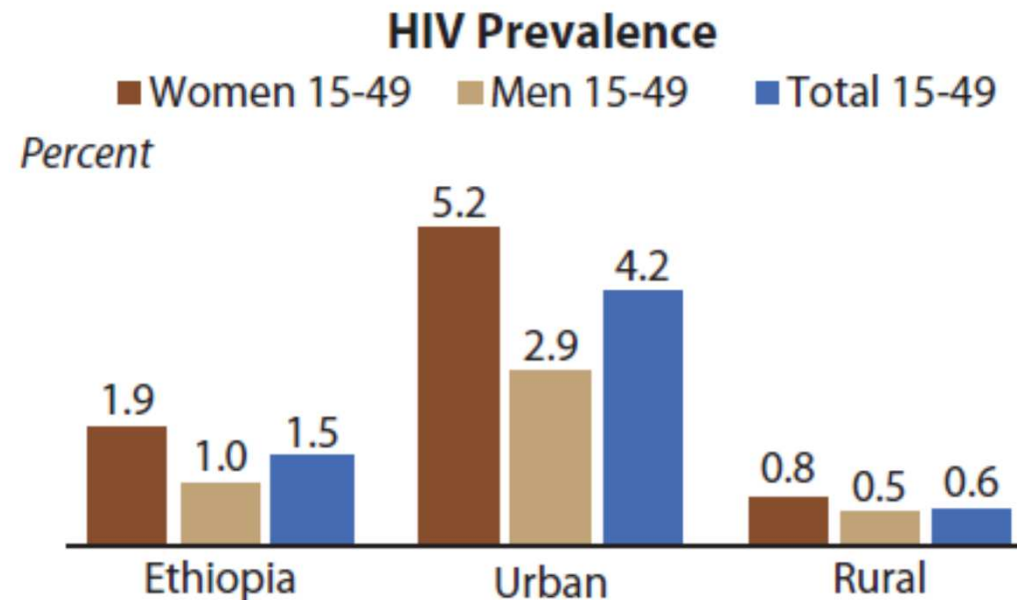


HIV/ADIS Cont...

- In 2019, among the estimated HIV-positive mothers that need ART for prevention of mother-to-child transmission (PMTCT), 74% received antiretroviral therapy (ART).
- PMTCT service is available in 84% of public health facilities, but **only in 5% of private facilities** (SARA 2018).
- In 2019, 64% of HIV-exposed infants born to HIV-positive women received virological testing for HIV;
 - ✓ but proportion of infants born to HIV-positive mothers who received antiretroviral prophylaxis is low

Major challenges and gaps in response to HIV epidemic in Ethiopia

- Suboptimal HIV case finding, especially in pediatric and adolescent age groups and in key and priority populations; and
- Gaps on overall quality of services for prevention, diagnosis, care, and treatment.



Tuberculosis (TB)

- Ethiopia is on track to achieving one of the three targets of the global End TB Strategy:
- TB incidence has declined by 21% from the 2015 estimate (was against a target of 20%).
- However, progress in reducing TB mortality was only 15% (target was 35%) under HSTP-I.
- As there is no nationally representative estimate of the catastrophic costs for TB-affected households, the status of progress on the third milestone is not known.

TB cont....

- TB case notification has been improving, with a detection rate of 71%.
- In 2019/20, TB treatment success =95% and cure rates also = 80%, (MOH, 2019)
- According to SARA 2018,
 - ✓ 50% of health facilities offered TB diagnosis by sputum smear microscopy examination and
 - ✓ 39% by clinical symptoms.
- Multi-drug-resistant (MDR) TB is a public health concern as the number of cases increase in Ethiopia.
- Since 2009, a cumulative of 4,906 drug-resistant TB (DR-TB) patients were detected and enrolled on second-line drug treatment.
- As of 2019, 59 DR-TB treatment initiative centers, and
- More than 700 treatment follow-up centers providing DR-TB treatment services.

	TB prevalence among HIV patients (%)	HIV prevalence among TB patients (%)
Addis Ababa	1.0	22.5
Afar	8.5	8.1
Amhara	0.6	8.2
Beneshangul Gumuz	7.7	6.3
Dire Dawa	1.0	12.6
Gambela	9.6	20.8
Harari	0.9	8.9
Oromiya	0.7	4.8
SNNP	1.6	3.8
Somali	14.5	5.5
Tigray	0.9	8.4
Ethiopia	0.8	7.4

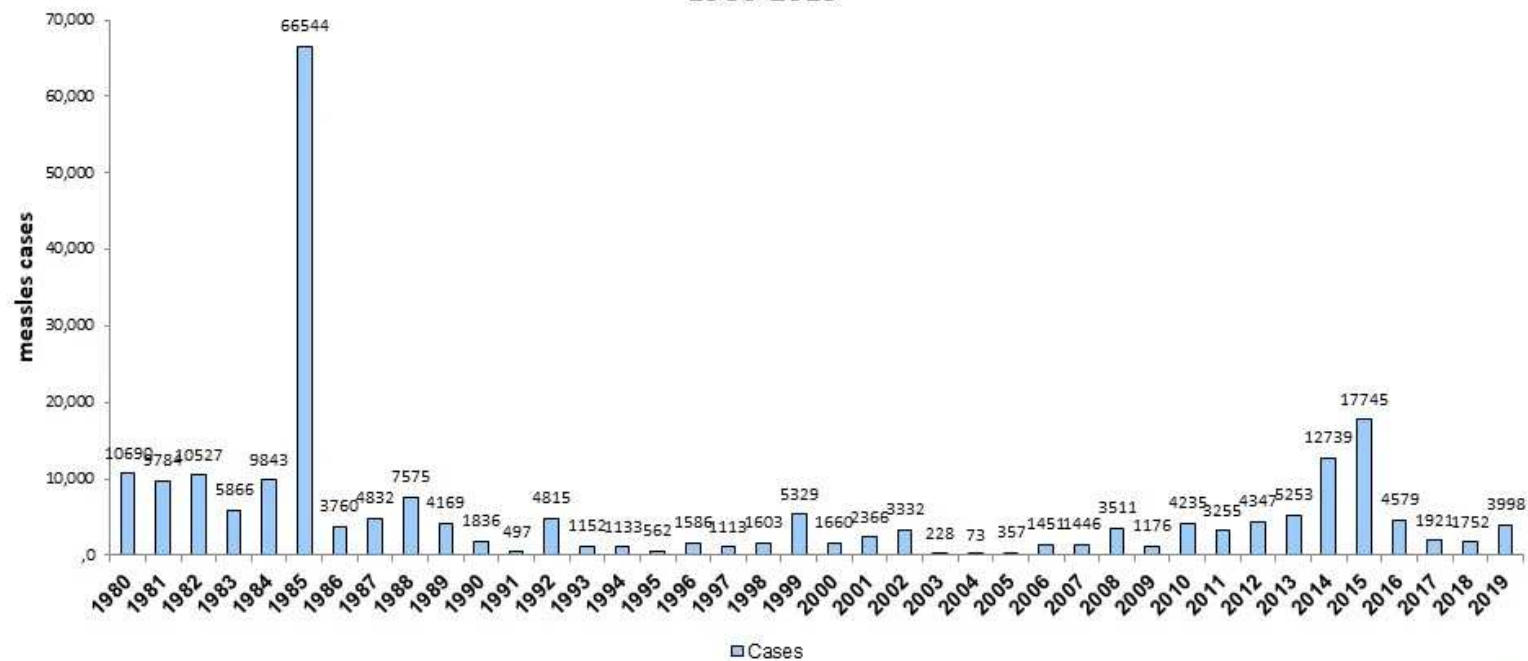
<https://doi.org/10.1371/journal.pone.0226127.t002>

Vaccine preventable disease(VPD)

- Various reports verified that the infant mortality rate is more common in low-income countries.
- Most children living in Sub Saharan Africa die every 12 months from communicable diseases that can be prevented through vaccination.
- For example, evidence of WHO from 2001 to 2008 revealed that around 9 million child deaths worldwide as a result of vaccine-preventable diseases,
 - ✓ a higher proportion occurred in sub-Saharan Africa, which was 4.4 million.
- In Ethiopia, vaccine preventable diseases contribute substantially to under-five mortality as well as morbidity.
 - ✓ **Diarrhea (18%), pneumonia (18%), measles (1%), and meningitis are the leading causes of child mortality in the country**

Number of reported Measles cases, Ethiopia 1980-2019

Date of chart: 12-Oct-2020



Number of reported cases.

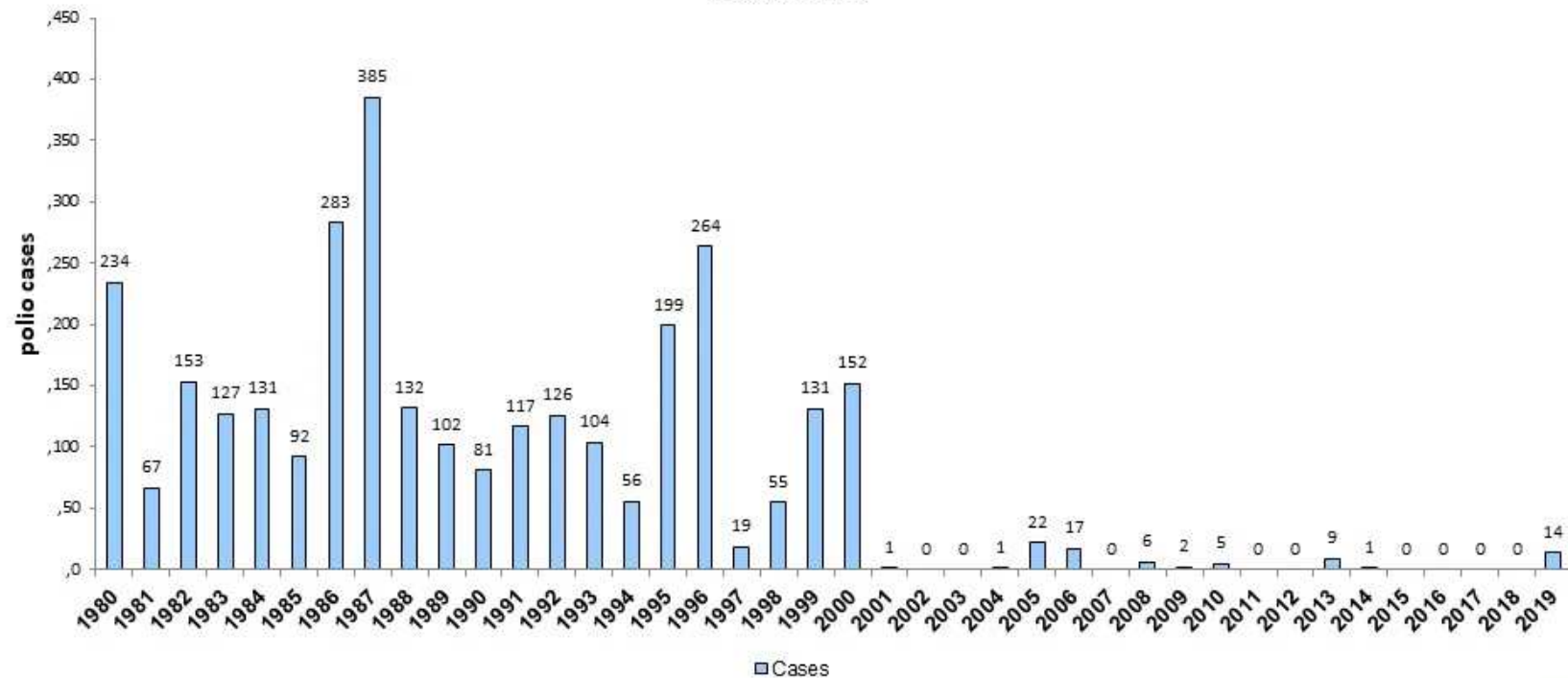
Source:

WHO/IVB database, data reported to WHO by Member States (http://apps.who.int/immunization_monitoring/globalsummary/timeseries/tsincidence measles.html) as of 12 October 2020.



Number of reported Polio cases, Ethiopia 1980-2019

Date of chart: 12-Oct-2020



Number of reported cases.

Source:

WHO/IVB database, data reported to WHO by Member States (http://apps.who.int/immunization_monitoring/globalsummary/timeseries/tsincidencepolio.html) as of 12 October 2020.

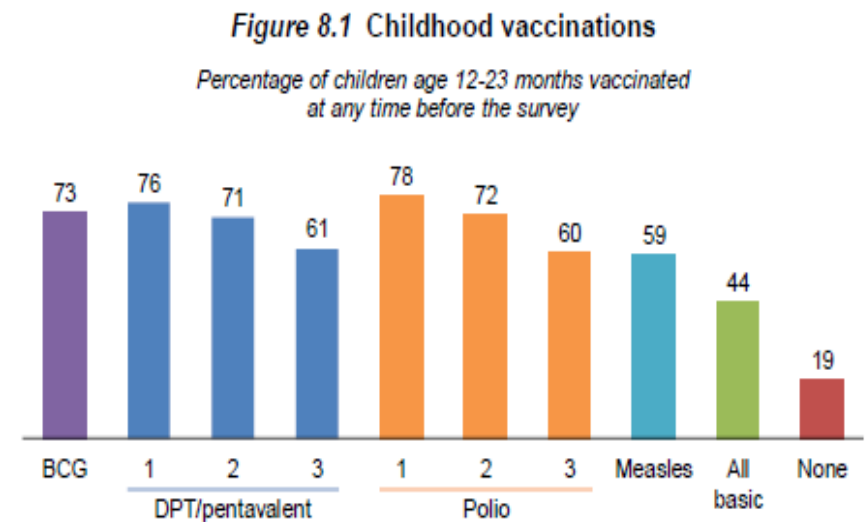


VPD cont....

- Ethiopia's EPI was launched in 1980, and the routine immunization schedule comprises six vaccine preventable diseases,
 - ✓ They are measles, diphtheria, pertussis, tetanus, polio, and tuberculosis
 - ✓ with the goal of increasing immunization coverage by 10% annually.
- Rotavirus, pneumococcal, and second dose of measles vaccines were introduced in 2011, 2013, and 2019 respectively

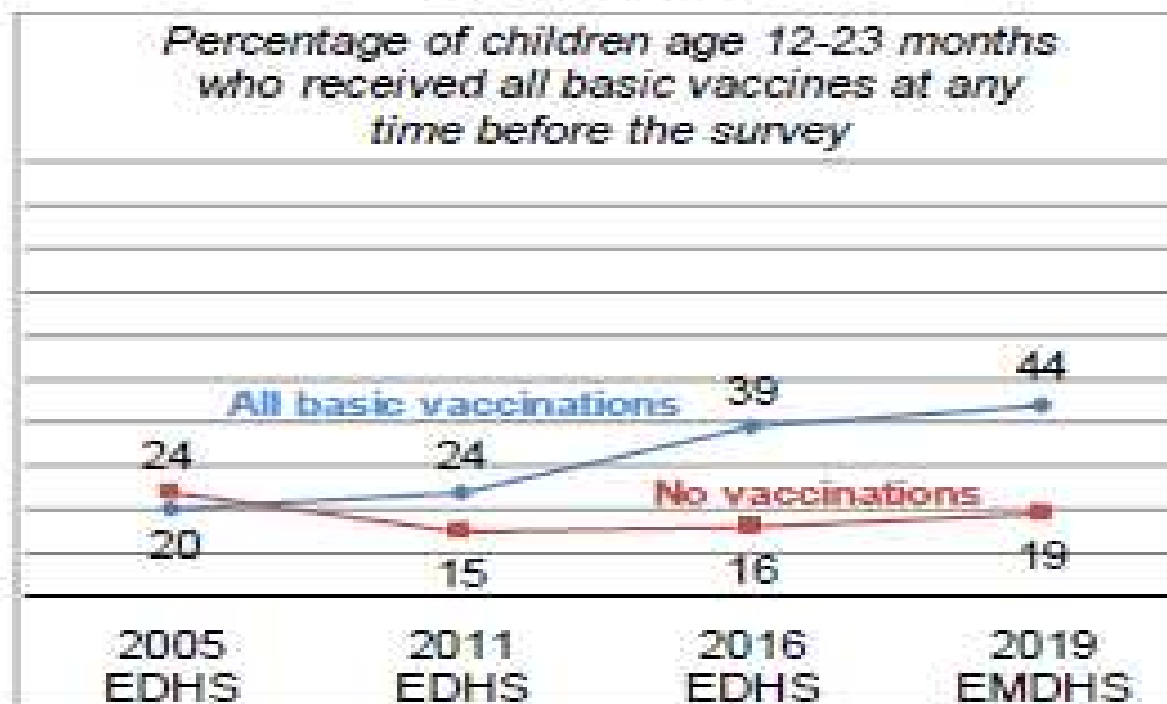
Immunization Status

- In 2019, coverage with three doses of pentavalent vaccine and all basic vaccines among children under 1 year of age reached **61%** and **44%**, respectively.
- However, about 19% of children had no vaccination at all.
- There is also high vaccination dropout rate with a national dropout of **13%** from pentavalent 1 to measles, with high regional variation.



Min EDHS 2019

Figure 8.2 Trends in childhood vaccinations



Min EDHS 2019

Sexually transmitted infections(STIs)

- More than **1 million** sexually transmitted infections (STIs) are acquired every day worldwide.
- Each year, there are an estimated **376 million new infections** with 1 of 4 STIs:
 - ✓ chlamydia, gonorrhoea, syphilis and trichomoniasis
- In 2016, WHO estimated 376 million new infections with 1 of 4 STIs:
 - ✓ Chlamydia (127 million),
 - ✓ Gonorrhoea (87 million),
 - ✓ Syphilis (6.3 million)
 - ✓ Trichomoniasis (156 million).
- About 80–90% of the global burden of STIs which found in the developing world where there is **limited and/or no access to diagnostics of STIs**

STIs cont...

- An estimated **240 million people** are living with chronic hepatitis B globally.
- Both HPV and hepatitis B infections are preventable with vaccination.
- A person can have an STI without having obvious symptoms of disease.

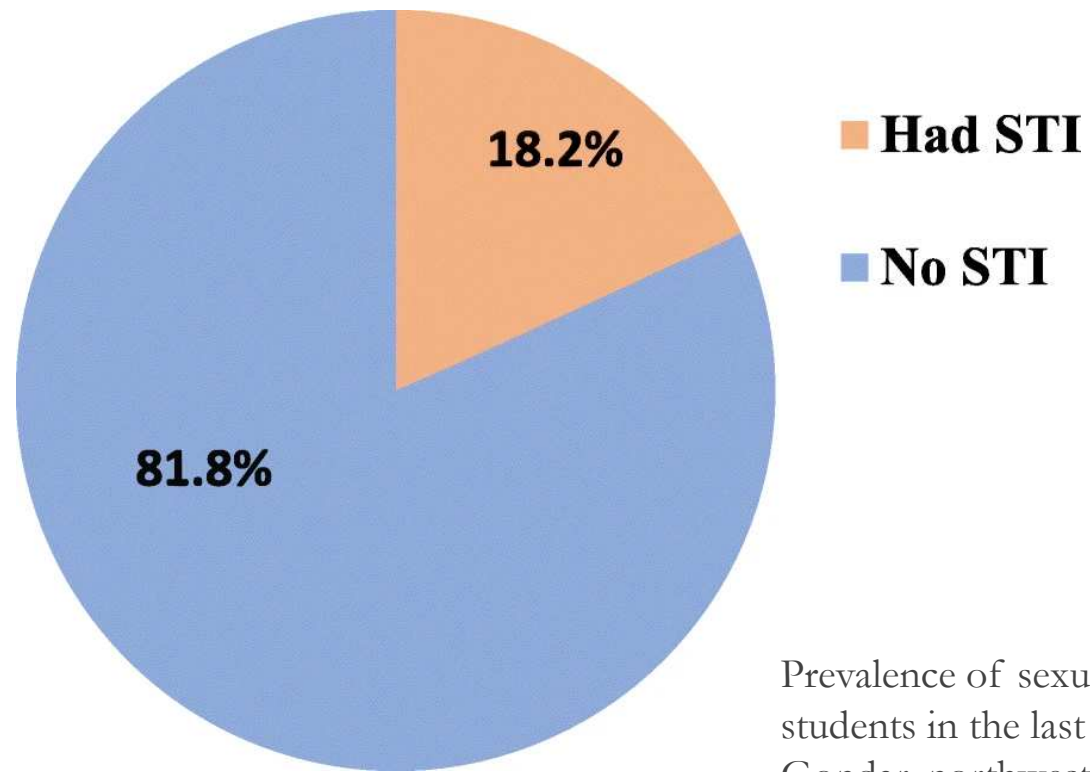
Common symptoms of STIs include;

- ✓ Vaginal discharge, urethral discharge or burning in men, genital ulcers, and abdominal pain.

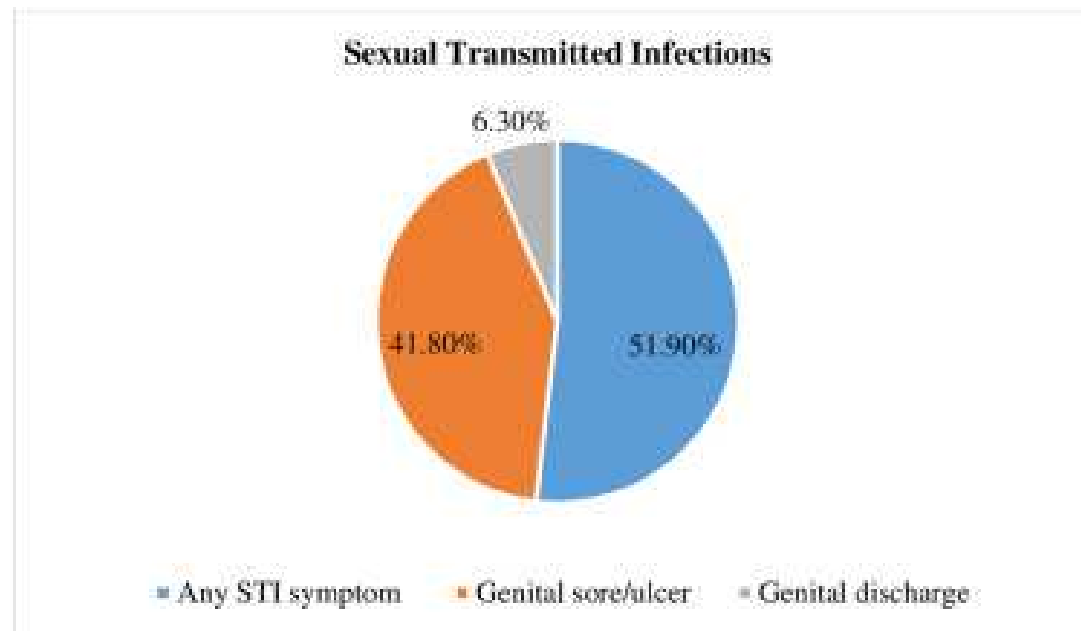
STIs cont....

In Ethiopia

- According to 2016 EDHS, Overall, 4% of women and men age 15-49 reported having an STI and/or symptoms of an STI in the past
- 2011 EHDS, reported 3% of women and 2% of men reported having an STI or STI symptom in the year before the survey.
- In Ethiopia, the highest reported rates of STIs are found among **15–24-years old**
- **67% of women and 66% men** did not seek any advice or treatment



Prevalence of sexually transmitted infections among students in the last 12 months, the University of Gondar, northwest Ethiopia, 2016



Percentage of self -reported STIs among women of reproductive age women in Ethiopia
<https://doi.org/10.1016/j.heliyon.2021.e07524>

Thank you!